PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MKALAMA DISTRICT COUNCIL

FORM THREE TERMINAL EXAMINATION MARKING SCHEME

032 CHEMISTRY

SECTION A (16 Marks)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
D	D	B	C	B	A	B	E	C	A

1mark@ = 10 marks

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i	ii	iii	iv	v	vi
C	B	A	I	F	D

1mark@ = 06 marks

3. **In laboratory**

- Reaction of dilute hydrochloric acid with Zinc metal **(1 ½ mark)**

$Zn_{(s)} + 2HCl_{(aq)} \longrightarrow ZnCl_{2(s)} + HCl_{2(g)}$ **(½ mark)**

In industry

- By steam reforming **($\frac{1}{2}$ mark)**

  Steam reacts with methane to produce carbon monoxide and hydrogen gas

- $H_2O_{(g)} + CH_{4(g)} \longrightarrow CO_{(g)} + 3H_{2(g)}$ **($\frac{1}{2}$ mark)**

b) Copper (ii) oxide is reduced to copper metal as shown below:-

$CuO_{(s)} + H_{2(g)} \longrightarrow Cu_{(2)} + H_2O_{(g)}$ **(1 mark)**

c) i. Hydrogen does not cause environmental pollution

ii. Hydrogen can be produced from renewable resources eg. Water

1mark@ = 02 marks

d)

Property	Uses
i) Reducing agent	Manufacturing margarine
ii) Lighter than air	Filling weather balloons
iii) Highly flammable	Preparation of water gas
iv) Readily react with other substances	Manufacturing of ammonia

1mark@ = 04marks

4.a) i. Data given

Mass of $M(OH)_2$, m= 5.8g

Molarity of HCl, M= $0.2M_{T.}$

- Required, atomic mass of metal M, Mr.
- Consider the following equation

$M(OH)_{2(aq)} + 2HCl_{(aq)} \longrightarrow MCl_{2(aq)} + 2H_2O_{(I)}$

1mol of $M(OH)_2$ ≡ 2mol of HCl

X? ≡ 0.2mol of HCl

$X = \frac{0.2mol}{2} = 0.1mol$ **00 $\frac{1}{2}$ mark**

Moles of $M(OH)_2$ = 0.1mol

From

Moles = $\frac{mass}{Molar\ mass}$

0.1mol = $\frac{5.8g}{}$

Mr

00½ mark

Mr. 58g/mol
Molar mass = 58g/mol

- In magnitude molar mass is equal to relative atomic mass

Hence
Atomic mass of M $(OH)_2$ = 58

$M(OH)_2$ = 58

M + 32 + 2 = 58 **00½ mark**

M= 58 – 34 =24

= 24

Atomic mass of M $(OH)_2$ = 24 **00½ mark**

ii. The name of metal M is magnesium **01 mark**

4b) Data given

Volume of acid , Va = 25cm3

Molarity of base, Mb = 1M **01 mark**

Molarity of acid, Ma = 1M

Consider the following equation

$X(OH)_{2(aq)} + 2HA_{(aq)} \longrightarrow XA_{2(aq)} + 2H_2O_{(I)}$ **01 mark**

na = 2

nb = 1

From

$\frac{Ma\ Va}{MbVb} = \frac{na}{nb}$ **01 mark**

$$\frac{1 \times 25}{1 \times vb} = \frac{2}{1}$$

$$\frac{2vb}{2} = \frac{25}{2}$$

$Vb = 12.5cm^3$

The volume capacity of the pipette used = $12.5cm^3$ **01 mark**

c) The burette should be cleaned and rinsed in order to avoid contaminations that may interfere with the reaction. **2 marks**

5 . a) water sample C is hardest, because it remains hard even after boiling.
2marks
b) water sample D behave as distilled water because it use Little volume of soap solution. **2 marks**
It means that water contains no soluble substances.
c) (i) Calcium hydrogen carbonate or magnesium hydrogen carbonate.
1marks
(ii) Calcium sulphate or magnesium sulphate **2 marks**
d) - using more soap during washing
- destruction of boilers, hot water pipes and kettles.

Any 3 points, 1 mark @ = 2marks

6. a) Laboratory rules should be obeyed by everyone because laboratory rules help;
- To have successful practical activities
- To ensure safety
- To protect equipments/apparatii in the laboratory
- To avoid contaminations **(Any 3points,1mark@=3marks)**

b) i. A laboratory should be equipped with fire extinguisher so as to extinguish harmful fire in case there is fire accident.
ii. A laboratory should have large windows for proper and good ventilation
iii. A laboratory should have fume chamber for preparing poisonous gases

(1mark@=3marks)

c) i. - In hospital
- In petrol filling station

- In chemical industries

(Any 3points, 00¹mark@=1 ¹_ *marks*)

ii. This is because safety signs ensure safety in the laboratory during practical activities. Hence safety signs warn laboratory users on the hazards of chemical and advice laboratory users to handle chemicals carefully

(1$\frac{1}{2}$ marks)

7. a) given data.

Mass = 0.65g

Molar mass = 98g/mol.

Volume = needed **½ marks**

Firstly, calculate number of moles then convert into volume

$n = \frac{\text{mass}}{\text{Molar mass}}$ **½ marks**

$n = \frac{0.65}{65.5g/mol}$

= 0.0099mol.

From the eqn.

Zn + 2HCl ------> ZnCl2 + H2 **½ marks**

This means

1mol.(Zn) -------> 1mol. (H)

0.0099 ----------> ?

? = 0.0099 x 1

1

= 0.0099mol(H) **½ marks**

Convert the obtained moles of hydrogen into volume.

From.

n = volume

Molar gas volume

Volume = n x molar gas volume **½ marks**

= 0.0099 x 22.4

= 0.222dm^3. **½ marks**

(b) given data.

Molarity = 0.1M

Volume = 400cm^3

Mass = needed

Molar mass = 98 **½ marks**

Firstly find the number of moles in 400cm^3 then convert moles into mass.
Remember that Molarity means the number of moles of solute that dissolved to make one litre
of the solution.
0.1mol. -------> $1000cm^3$
? <---------- $400\ cm^3$
?= 0.1 x 400
1000
0.04mol(in $400cm^3$) **½ marks**
Convert 0.04mol. Into mass.

$$n = \frac{\text{mass}}{\text{Molar mass}}$$

Mass = n x molar mass
= 0.04 x 98
= 3.92g. **½ marks**
(c) (i) data given.
Volume = $100cm^3$
Mass = 5.3g
Molarity = required
Molar mass = 106g/mol. **½ marks**
But molarity = number of mole in one litre of the solution.
Also

$$molarity = \frac{\text{concentration(g/dm3)}}{\text{Molar mass g/mol}}$$ **½ marks**

Convert 5.3g in 100cm^3 into g/L(litre)
5.3g ---------> $100cm^3$
? <---------- $1000cm^3$
? = 5.3 x 1000
100
= $53g/dm^3$ **½ marks**

$$molarity = \frac{\text{concentration(g/dm3)}}{\text{Molar mass g/mol}}$$

$$molarity\ \frac{53(g/dm3)}{106(\frac{g}{mol})}$$

= 0.5M **½ marks**
(ii)
data given.
Volume = 2L ($2000cm^3$)

Mass = 80g
Molarity = required
Molar mass = 40g/mol. **½ marks**
But molarity = number of moles in one litre of the solution.
Also =

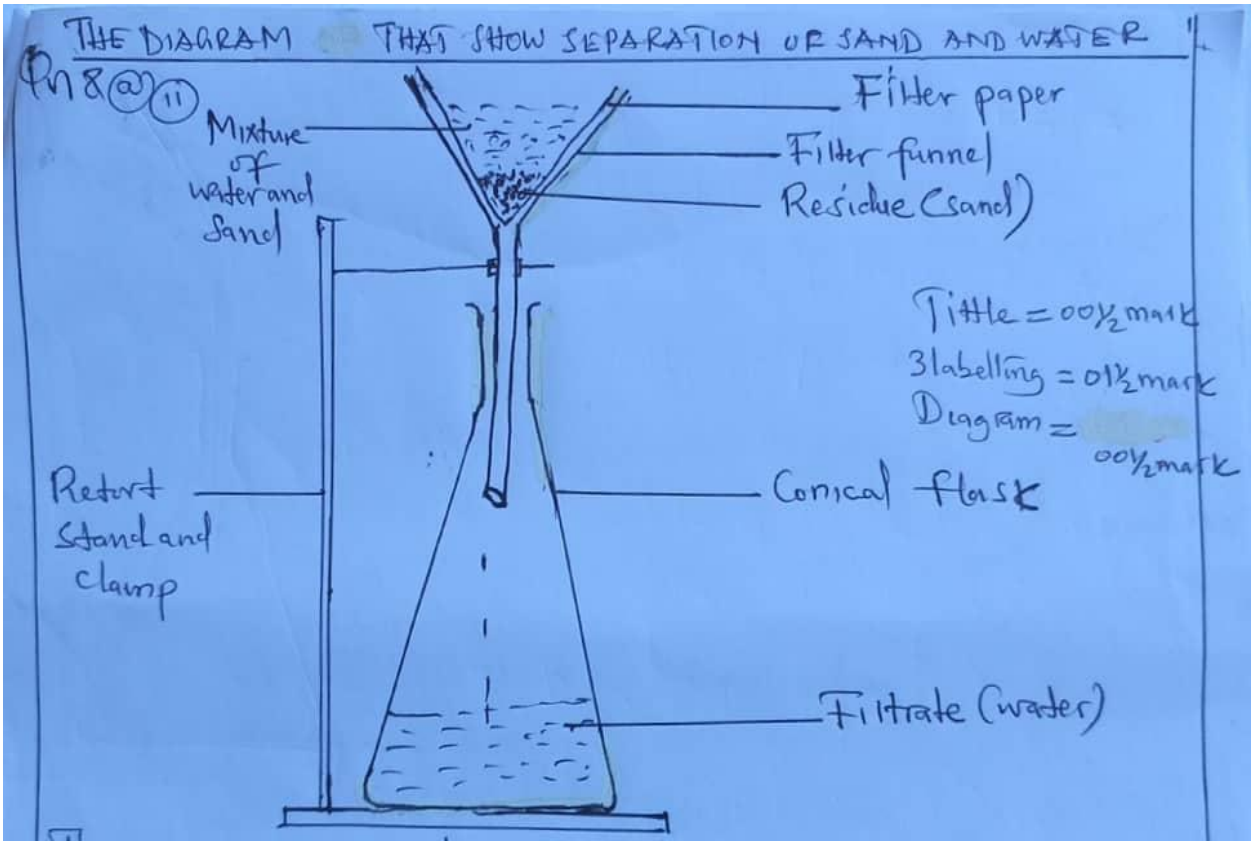
$$molarity = \frac{\text{concentration(g/dm3)}}{\text{Molar mass g/mol}}$$

s Convert 80g in 2000cm^3 into g/L(litre) or dm^3
80g ---------> 2000cm^3
? <---------- 1000cm^3
? = 80 x 1000
2000
= 40g/dm^3 **½ marks**
26
Molarity = concentration
Molar mass
= 40
40
= 1M **½ marks**

Total 8 marks

8a) i. Conical flask/Flat = bottomed flask

- Filter funnel
- Retort stand and clamp
- Filter paper **(Any 3points, ^{1}mark@=1 ^{1}marks)**
  2 **2**



The sand is separated from the liquid using a porous filter such as filter paper. The sand will remain in the filter paper as residue and water will pass through filter into the conical flask.

00 $\frac{1}{2}$Mark

8b) - Solution ⟶ used in cleaning eg. Soap solution

- Suspensions ⟶ Used as drug for treatment of diseases
- Emulsion ⟶ Used in decoration eg. Paint

(– ½ mark@=1½marks)

c) - Used in drying crops
- Used in drying clothes
- Used in drying mopped areas
- Used in drying washed hairs

Any 3points, 1mark@=03marks

SECTION C: (30 Marks)

9.a) This is because the antacids will neutralize excess acid in the stomach, hence the problem of excess acid will be overcome. **02 marks**

b) $Mg(OH)_{2(aq)} + 2HCl_{(aq)} \longrightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$ **02marks**

c) Data given

Normal acid content = 160 millmolar

Abnormal acid content = 210millimolar

Excess stomach acid = 210 – 160

Excess acid = 50millimolar

Converting millimolar to molarity

1mM ≡ $0.001mol/dm^3$

50mM ≡ X?

$$X?^{3} = \frac{50 \times 0.001 mol/dm}{1}$$

$$= 0.05M$$

- Molarity of excess acid = 0.05M 01mark

- Determining number of moles of excess acid,

$$\text{From Molarity} = \frac{\text{moles (n)}}{\text{Volume (dm3)}}$$ $00\frac{1}{2}$mark

$$0.05M = \frac{\text{Moles(n)}}{0.5\text{litres}}$$

Moles = 0.025mol

Moles of excess acid = 0.025mol $00\frac{1}{2}$mark

- Determine the number of moles of antacid that will neutralize excess acid - consider the following equation.

$Mg(OH)_{2(aq)} + 2HCl_{(aq)} \longrightarrow MgCl_{2(aq)} + 2H_2O$

1mol of $Mg(OH)_2$ ≡ 2mols of HCl $00\frac{1}{2}$mark

X? ≡ 0.025mol of HCl

$$X = \frac{0.025 \times 1}{}$$

= 0.0125mol (00½mark)

Moles of antacid that will neutralize excess acid = 0.0125mol

- 0.0125mol of Mg (OH)2 will neutralize 0.025mol of HCl

Then

$$\text{Moles} = \frac{\text{Mass}}{\text{Molar mass}}$$

$$0.0125 = \frac{\text{Mass}}{58} \quad (1\text{mark})$$

$$\text{Mass} = 0.725\text{g}$$

 Mass of tablet to be given is 0.725g (1mark)

d) Mass of tablet = 0.725g = 725mg

1tablet ≡145mg (2mark)

X? ≡725mg

$$X = \frac{1\text{tablet} \times 725\text{mg}}{145\text{mg}}$$

= 5tablets (2mark)

 The student will be given 5 tablets.

e) 2½ days (02mark)

10a) i. Neutralization reaction (1mark)

ii. – Used in relieving indigestion

- Used in controlling soil PH
- Used in treating factory wastes **(Any 4points,1mark@=4marks)**

- Preventing formation of acid rain
- Manufacturing of fertilizers

b) i. Lead (II) sulphate

It is prepared by precipitation reaction (1mark)

$Pb(NO_3)_{2(aq)} + Na_2SO_{4(aq)} \rightarrow PbSO_{4(s)} + 2NaNO_{3(aq)}$ (1mark)

ii. Copper (II) nitrate

It is prepared by reacting dilute HNO_3 with metal carbonate

(1mark)

$CuCO_{3(s)} + 2HNO_{3(aq)} \longrightarrow Cu(NO_3)_{2(aq)} + CO_{2(g)} + H_2O_{(s)}$

(01mark)

iii. Calcium Chloride

It is prepared by reacting dilute HCl with metal carbonate (01mark)

$CaCO_{3(s)} + 2HCl_{(aq)} \longrightarrow CaCl_{2(aq)} + H_2O_{(q)} + CO_{2(g)}$ (01mark)

iv. Calcium Carbonate

It is prepared by precipitation reaction (01mark)

$Ca(NO_3)_{2(aq)} + Na_2CO_{3(aq)} \longrightarrow CaCO_{3(s)} + 2NaNO_{3(aq)}$ (01mark)

v. Barium sulphate

It is prepared by precipitation reaction (01mark)

$BaCl_2(aq) + Na_2SO_4(aq) \longrightarrow BaSO_4(aq) + 2NaCl(aq)$ (01mark)

11. i. Introduction

- Meaning of chemistry ($1\frac{1}{2}$mark)
- Meaning of Agriculture ($1\frac{1}{2}$mark)

ii. Main body

- It helps to make fertilizers
- It helps to produce pesticides **(Any 5points,2mark@=10marks)**
- It helps to produce animal vaccine

- It helps to produce weed killers
- It helps to make insecticides

iii. Conclusion (2mark)